



MINING DEALER BEST PRACTICE

Cooling System Thermal Inspection

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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**January 2015
1411-1.01-1359**

1.0 Introduction

This Best Practice aims to establish a preventive maintenance procedure to help forewarn unscheduled downtime, reduced PCR or even catastrophic engine or converter failures due to high operating temperatures.

Efficiency in the cooling system decreases when external and/or internal obstruction of radiator cores happen. This produces high engine and/or converter operating temperatures.

An effective way to determine if cores present internal/external obstruction is by using thermography.

Thermographic inspection as part of maintenance is a strategy to:

- Determine possible internal and/or external clogging of radiator cores
- Avoid non-scheduled downtimes due to engine or converter overheating
- Avoid catastrophic failures related to high temperature
- Avoid excessive repair costs of severe wear components due to high operating temperatures
- Reduce assessment time over engine temperature. Prevents customer discomfort
- More graphical diagnosis, easier to understand, quick define and very technical

2.0 Best Practice Description

The inspection process is mainly focused on using a thermal imager or camera (417-5122 or other similar or with best features) and can be replicated by other dealers worldwide.

When executing thermographic inspection, the following considerations must be taken into account:

- Inspection must be performed in the field
- Engine temperature must be over 90°C (Work the machine / engine to get all systems hot)
- Full engine rpm's
- Avoid reflected high temperatures
- Consider emissivity equal to 0.95 for pipes (painted)
- Assessment for radiator's internal clogging will be performed qualitatively (comparative)
- Cooler deltas' assessment will be performed quantitatively

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I. Inspection Procedure for Radiator's Condition (qualitative analysis):

Radiator's internal obstruction assessment is performed due to comparison (qualitative)

- Ask Operations for authorization to perform tests with equipment on the field
- Head to equipment and coordinate with operator:
 - ✓ Park equipment
 - ✓ Accelerate
- Ensure machine is safe before approaching
- Stand behind equipment, approximately 5-6m distance (see Fig1 & Fig2)
- Take thermal image (See Fig3 & Fig4)



Fig1. Location to register thermal image



Fig2. All radiator must be captured completely on the thermal image area

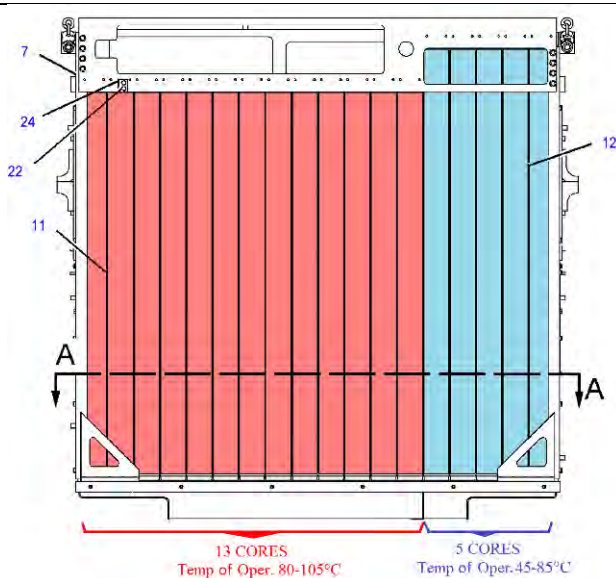


Fig3. Radiator cores disposition

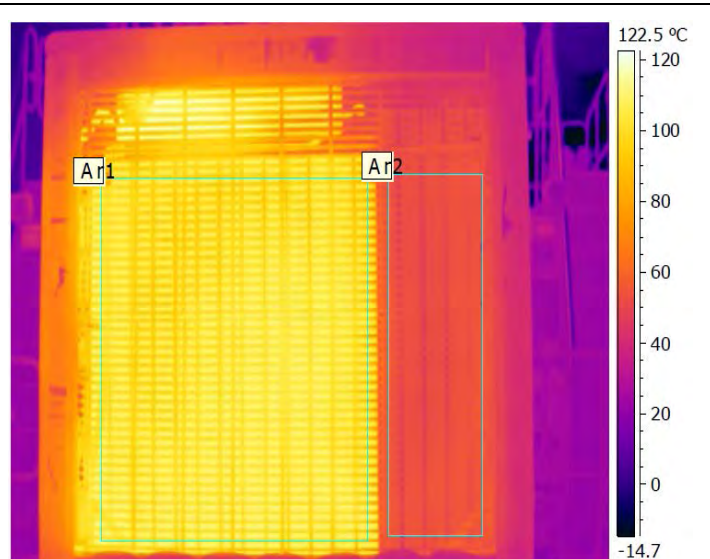


Fig4. Example of how to take a thermal image. Radiator in good conditions

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II. Assessment Procedure of radiator's delta temperature (quantitative):

This assessment's objective is to measure coolers' efficiency by using delta temperature and quantitative analysis from thermal image.

- Climb machinery through its left side and record temperatures of the following parts:

System	Temp Location (On Diagram)	Referenced Value	Measured Value	Expected Temperature
Radiator Inlet	T1	107.9		Max. 107°C
Radiator Outlet	T2	99.3		
Radiator Delta	T2-T1	-8.4		Min. Δ 8°C

Chart 1-Checkpoints for radiators' delta

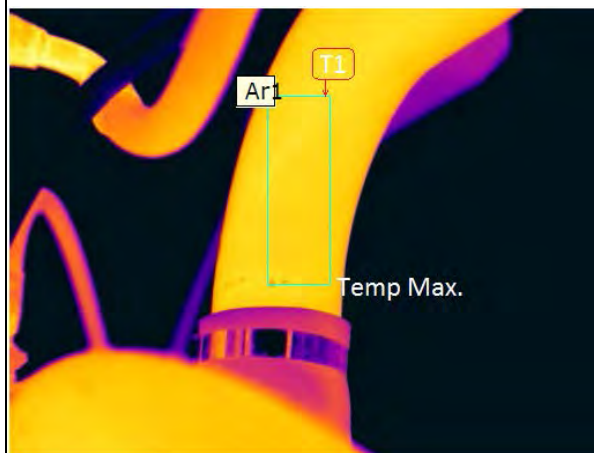
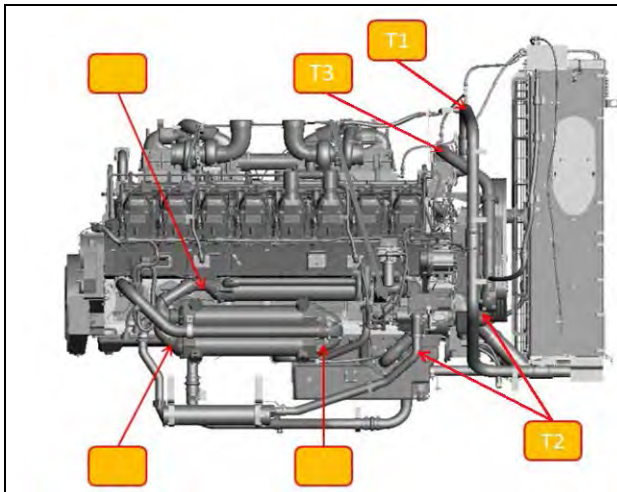


Fig5. (T1) Radiators' inlet temperature

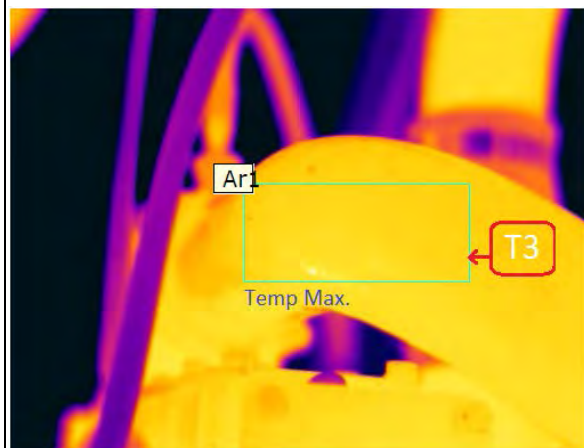


Fig6. (T3) Temperature towards the water pump (referential temperature)

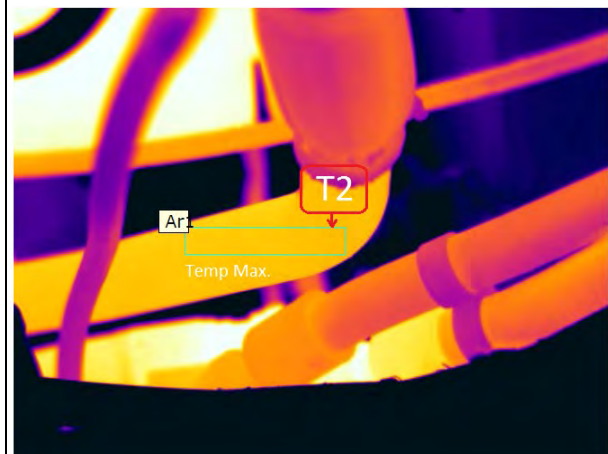


Fig7. (T2) Radiators' outlet temperature

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III. Thermal Images Analysis:

1. To assess the radiator's condition (internal obstruction or clogging) compare all cores with another, heat distribution must be uniform among them all. As example, Fig4. shows a uniform heating distribution in all cores
- Thermal images that DO NOT show uniform distribution are shown below. These do have "cold zones" which is clear evidence that there exists internal obstruction (Fig9 & Fig10)

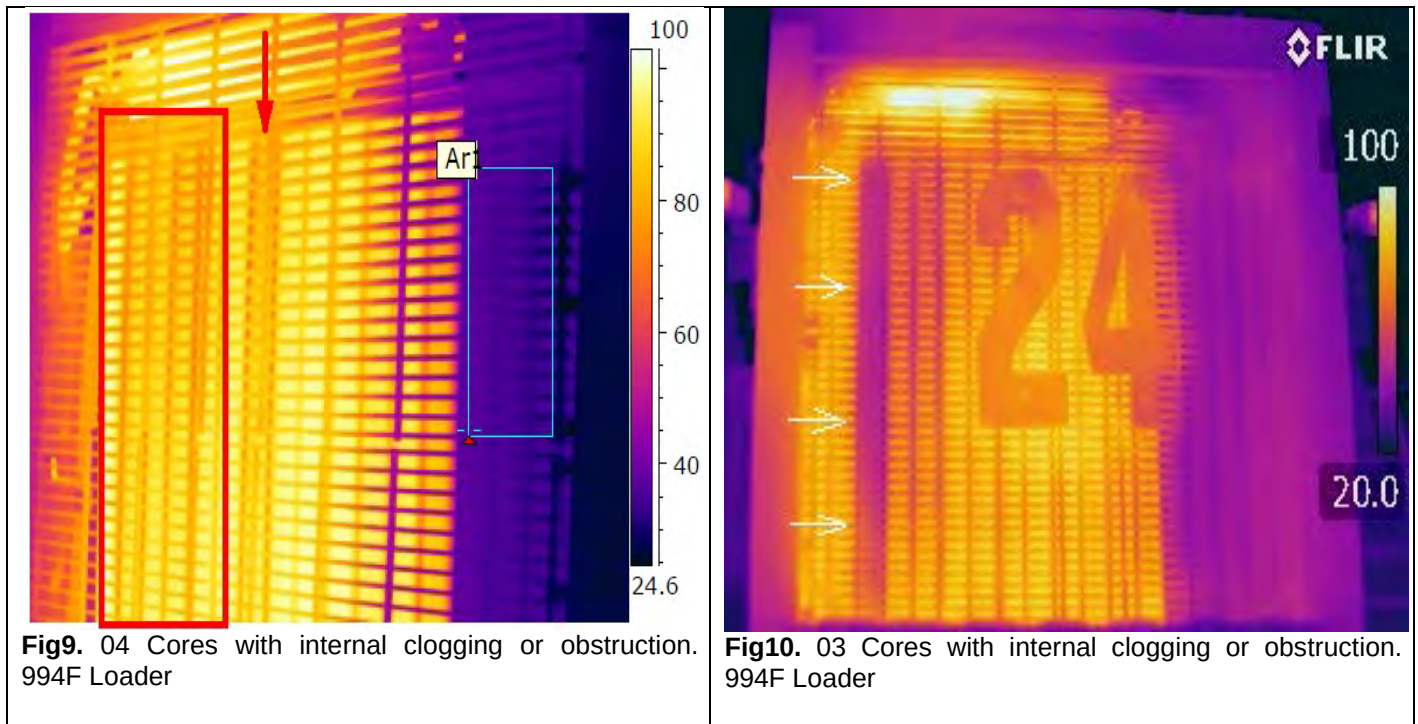


Figure 6 (Fig6.) shows an inspection performed on our customer's 994F loader in Cerro Verde, where 03 cores were clogged (15% of the total) generating engine-over-heating. After cores were replaced, the issue was fixed.

Figure 5 (Fig5.) shows 04 clogged cores on our customer's 994F. After cores replacement, the engine-over-heating issue was fixed.

- Schedule cores replacement in case internal clogging is found
2. In case there is no internal clogging, analyze temperature delta of thermal images from T1 & T2 points (see Chart 1) – Fig5, Fig6 & Fig7
- If delta temperature is lower than indicated in chart 1, it may indicate that external clogging of radiator or hydraulic cooler is happening. Set the radiator's cores external cleaning function (use air or hot water)

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3.0 Implementation Steps

This Best Practice arose as a necessity to identify possible internal clogging of radiator cores. Two failures we found due to radiator's internal clogging in a customer's mine, where the failure determination process led to a quite long assessment period.

This Best Practice can be replicable by other dealers worldwide.

Implementation Steps are:

- Assessment must be scheduled in maintenance planning
- It is recommended to have a 2000hrs periodicity
- Set who will be responsible of data compilation
- Coordinate with the person in charge of machine operation to have access to the equipment
- Coordinate with operator a 15 minute downtime with the equipment turned on
- Ensure machine is safe to approach
- Quickly take thermal images to the radiator from the equipment's rear-end
- Complete pre checks
 - I. Check for proper coolant level
 - II. Check for debris build-up in airstream of all cooling cores
 - III. Check for coolant leaks
- Assess thermal images, record them, schedule cores' replacement as needed and issue a report

4.0 Benefits

Efficiency loss in the engine's cooling system ends up as high operating temperatures both, engine and torque converter, causing high internal wear or catastrophic failure as a result of overheating.

In one example from Southern Peru, 4100masl, 03 catastrophic failures of 3516B engines generated a very high loss. New engines were purchased for US\$570,253.50.

Implementing periodic thermographic inspections as strategy ensures the engine and converter to reaches their PCR. The torque converter in the mine was repaired on average every 5000hrs of operation, the cost of repairing it is US\$48,886.00. This cost can be eliminated by implementing this Best Practice.

Powertrain overheating issues cause unscheduled equipment downtime. This means that the equipment's availability is lower than expected. Cerro Verde mine 994F loader's availability decreased to 80% due to overheating issues. But after implementing thermographic inspections and repairing the radiator, loader's availability remains above 88%.

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Benefits of implementing this Best Practice are:

- Avoid catastrophic failures
- The component reaches its PCR
- Reduces rebuild cost
- Reduces rebuild time
- Reduces down time
- More productivity
- Better availability
- Quick define, graphic and easier to understand

5.0 Resources Required

Investment needed to implement this Best Practice:

- Trained staff on thermal cameras
- Pick-up truck to transport staff to where the front loader is located
- Fluke Ti32 Thermal Camera 417-5122 or Flir E60 or more advanced thermal cameras

6.0 Supporting Attachments/References

- SEBD0518-10 Know Your Cooling System {0708, 1000, 1350, 7000}.

7.0 Related Best Practices

- Best Practice: “Cooling system Performance for LWL 994”

8.0 Acknowledgement

This Best Practice was written by:

Fidel Sarmiento Córdova
 Auxiliary Fleet Support Engineer/Technical Department
fidel.sarmiento@ferreyros.com.pe
 511 984058638

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